# Time Refinement and an Action Scale

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**Abstract**

Free-particle kernels give an exact test of refinement-based proposals for an action scale. Integrating over an inserted position preserves a Gaussian family; time homogeneity makes its variance linear in duration. The proportionality constant remains a model parameter. Fixed-endpoint bridges concentrate on the Newtonian straight path as that parameter vanishes. We use these standard calculations to isolate the physical selection question in the proposed connection between geometric subdivision, renormalisation and quantum mechanics.

## 1 A source-driven experiment

The experiment is to refine a time interval, integrate out the inserted positions, and track the remaining parameters. Rivero's cone discussion [5] motivates retaining rescaled data as sections approach; his path-integral proposal [4] motivates a finite blocking map. Brouder's rooted-tree correspondence [1] supplies an algebraic route to nonlinear composition, and the tangent-groupoid construction [2] supplies a separate quantum-completion reading.

This note works on the free line with mass $m > 0$. Positions have units of length, $s, t, T > 0$ are durations, and $\kappa$ has units of action. An identification $\kappa = \hbar = h/(2\pi)$ is a physical interpretation of a selected oscillatory model. The present calculations classify a free family; the selection of its physical member is the open reconstruction question.

## 2 Eliminating an inserted position

The normalized positive kernel is

$$G_t^\kappa(x) = \sqrt{\frac{m}{2\pi\kappa t}} \exp\left(-\frac{mx^2}{2\kappa t}\right), \qquad \kappa > 0. \tag{1}$$

It is a probability density on $\mathbb{R}$ with variance $\kappa t/m$; the standard Brownian normalization is given in [3, §4.1, (23)–(25)]. Its exponent is minus the free straight-segment action divided by $\kappa$. Here positivity defines a probability model; the oscillatory model enters in Section 4.

**Proposition 1** (Exact blocking)**.** *For $s, t, \kappa > 0$ and $x, z \in \mathbb{R}$,*

$$\int_{\mathbb{R}} G_t^\kappa(z-y) G_s^\kappa(y-x)\, dy = G_{s+t}^\kappa(z-x).$$

*Proof.* Set $\mu = (tx + sz)/(s+t)$. Completing the square gives

$$\frac{(y-x)^2}{s} + \frac{(z-y)^2}{t} = \frac{(z-x)^2}{s+t} + \frac{s+t}{st}(y-\mu)^2.$$

The remaining Gaussian integral is $\sqrt{2\pi\kappa st/[m(s+t)]}$. Multiplication by the two prefactors in (1) gives exactly the prefactor at duration $s+t$. $\square$

Consequently, for any finite partition $0 = t_0 < \cdots < t_n = T$, integrating the product of the $n$ kernels over its $n-1$ internal positions gives $G_T^\kappa(z-x)$. The action parameter is unchanged at every finite refinement; this identity requires no continuum path measure.

**Proposition 2** (The Gaussian parameter left by composition)**.** *Let $(\mu_t)_{t\geq 0}$ be a weakly continuous convolution semigroup of centered Gaussian probability measures on $\mathbb{R}$, allowing a degenerate Gaussian, with $\mu_0 = \delta_0$. Then $\mu_t$ has variance $w(t) = at$ for one $a \geq 0$. Conversely every $a \geq 0$ defines such a family.*

*Proof.* Characteristic functions are $\widehat{\mu}_t(k) = \exp[-w(t)k^2/2]$. Composition implies $w(s+t) = w(s) + w(t)$. Weak continuity, evaluated at $k = 1$, makes $w$ continuous. Additivity first yields $w(r) = rw(1)$ for nonnegative rational $r$, and continuity extends this to every real $t \geq 0$. Nonnegative variance gives $a = w(1) \geq 0$. These characteristic functions verify the converse. □

At fixed mass set $\kappa = ma$; since $[a] = \text{length}^2/\text{time}$, $[\kappa] = \text{action}$. All $\kappa > 0$ satisfy the same composition law, and $a = 0$ supplies the identity convolution family. Positive-width densities select $a > 0$ as an additional premise. Even that subclass has $\inf\{\kappa : \kappa > 0\} = 0$: positivity of each member and a uniform lower bound over a class are distinct questions.

## 3 A bridge to the Newtonian path

Conditioning on endpoints gives an exact trajectory comparison. The standard bridge construction is given in [3, §3.4]. For $0 < s < T$, define the normalized intermediate-position density

$$\rho_s^\kappa(y \mid x, z) = \frac{G_{T-s}^\kappa(z-y)G_s^\kappa(y-x)}{G_T^\kappa(z-x)}.$$

Proposition 1 makes its integral one. The same square completion gives mean and variance

$$q_{\mathrm{cl}}(s) = x + \frac{s}{T}(z-x), \qquad \sigma_s^2 = \frac{\kappa s(T-s)}{mT}. \tag{2}$$

The mean is the unique free Newtonian trajectory with these endpoints. For any displacement tolerance $r > 0$, Chebyshev's inequality gives

$$\Pr\{|Y_s - q_{\mathrm{cl}}(s)| \geq r\} \leq \frac{\kappa s(T-s)}{mTr^2} \longrightarrow 0.$$

For any fixed finite set of internal times, the product-kernel bridge has these marginals, and a union bound proves joint concentration on the sampled straight path. This is a finite-dimensional classical-path limit in a positive-measure model. Quantum phase-space correspondence is a further limiting problem.

The zero-variance convolution family describes a stationary position. General free classical motion instead has the phase-space transition measure

$$P_t((q,p), dq'\,dp') = \delta_{q+pt/m}(dq')\,\delta_p(dp'),$$

whose composition follows by substitution. This keeps the deterministic comparison on its appropriate state space.

## 4 The oscillatory family and its coefficient

For each $\kappa > 0$, the free oscillatory kernel is

$$K_t^\kappa(x) = \sqrt{\frac{m}{2\pi i\kappa t}} \exp\left(\frac{imx^2}{2\kappa t}\right), \qquad t > 0.$$

Use the unitary transform $\mathcal{F}f(k) = (2\pi)^{-1/2} \int_{\mathbb{R}} e^{-ikx} f(x)\, dx$. The kernel's precise meaning is the tempered-distribution kernel of the Fourier multiplier $\exp[-i\kappa t k^2/(2m)]$, with the square-root branch fixed by continuation from positive real Gaussian variance. On $L^2(\mathbb{R})$, these multipliers define a strongly continuous unitary group for real $t$: the group law follows by multiplication, unitarity by modulus one, and strong continuity by dominated convergence. Its Hamiltonian in the convention $i\kappa\partial_t\psi = H_\kappa\psi$ is

$$H_\kappa = -\frac{\kappa^2}{2m}\frac{d^2}{dx^2}, \qquad D(H_\kappa) = H^2(\mathbb{R}).$$

These are the usual free-particle formulas [6, §7.3]. Operator composition defines the oscillatory convolution, avoiding an unqualified interchange of conditionally convergent integrals.

Writing the positive family as $\exp[t\kappa\partial_x^2/(2m)]$ and continuing $t$ to $it$ yields this unitary family. That continuation is a specified model choice. Both constructions retain a free coefficient with action units.

## 5 The selection problem now exposed

The first task is to find a physical principle that selects a positive action scale and explains its universality under composition of interacting systems. Three parameters stay separate: temporal mesh size, the action scale $\kappa$, and any finite-speed parameter $c^{-1}$. The present free nonrelativistic model allows arbitrarily fine temporal meshes at each fixed $\kappa$.

The next source-guided experiment audits the two regulators in [4], starting with a finite-dimensional quadratic action and its exact normalization. A second task extracts the hypotheses and limiting topology of [2]. Together they test whether refinement consistency constructs a quantum completion, and which additional premise selects its physical action parameter. Interactions and finite propagation then become explicit changes of model.

## References


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